

FACTORS AFFECTING ERYTHROPOIESIS

3) Food factors:-

a) Proteins: proteins specially the essential amino acids are more effective in erythropoiesis.

b) Trace elements:-

1- Iron: It forms about 0.334% of the Hb molecule.

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Absorption and transportation of the iron:-

Food iron is ingested as $\text{Fe}(\text{OH})_3 \rightarrow$ free Fe^{+3} (by gastric HCl) $\rightarrow$ reduced to Fe^{+2} (by Vit. C & Ca^{+2}), which absorbed by the intestinal cells, then oxidized to $\rightarrow \text{Fe}^{+3}$ Apoferritin $\rightarrow$ **Ferritin** (stored form in the intestinal cells, liver, spleen and bone marrow). It can be reduced again to ferrous form and transported from the intestinal cell to the blood by a β - globulin (produced by the liver, called **transferrin**), to the bone marrow.

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Fe^{+2} form (not Fe^{+3} form) can cross the cell membranes. Phosphates and tannic acid (in **boiling tea**) forms insoluble iron which prevent its absorption.

Iron deficiency:-

It causes **microcytic hypochromic** anemia, which characterized by small size RBC and decreased Hb content. It is more common in **females** (due to cyclic bleeding, pregnancy and lactation) and **young's** (due to the mother's milk is deficient in iron).

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2&3) Copper(Cu) and Cobalt (Co):-

These elements acts as **catalysis** for **Hb synthesis** but do not form a part of Hb molecule.

Cu is carried in blood by α - globulin called **ceruloplasmin**. Cu deficiency leads to anemia.

Co stimulates erythropoietin formation and inter in the formation of vit. B12 (about 4% of its structure). Co is a dietary essential for **ruminants** because is needed in the ruminal bacterial **synthesis of vitamin B12**.

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c) Vitamins:-

1&2) Folic acid and Vitamin B12 (Cyanocobalamin):-

They are called **maturing factors**. They are needed for the formation of the nucleus of the early RBC development.

Vit. B12 present in the food (called extrinsic factor) combined with protein. Gastric Hcl freed it from protein then, gastric enzyme called intrinsic factor help its absorption by the ileum, then transfer to the blood and stored in the liver (as a **hematinic principle**) till used by the bone marrow.

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Deficiency of Vit.B₁₂ leads to pernicious anemia, so it is known as **antipernicious anemia** factor. This type of anemia rarely occurs in ruminant animals because the Vit.B₁₂ is synthesis by its ruminal microflora.

Folic acid is converted to an active form **folinic acid** in the body and absorbed in the **jejunum**. It is like Vit. B₁₂ in its action but cannot improve the nervous lesions of pernicious anemia.

3- Vitamin C:-

It **stimulated the erythropoiesis** through stimulation of tissue growth and metabolism of the body including the bone marrow; also it helps the absorption of iron.

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4) Hormones:-

- a) **Erythropoietin hormone:** It is the main hormone that stimulates the bone marrow for erythropoiesis.
- b) **Thyroxine:** It is the most powerful stimulant of the body metabolism and energy production, so it is very important for erythropoiesis.
- c) **Testosterone:** It stimulates erythropoiesis through increasing secretion of erythropoietin hormone (this may explain why RBCs count is higher in male than female).

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5) Liver:-

Healthy liver is essential for normal erythropoiesis because:-

- 1- Manufacture of **globin of Hb**, **α - globulin** (which carry the erythropoietic factor), **β - globulin** (which carry the iron) and **ceruloplasmin** (which carry the copper).
- 2- Chair in the production of about **15% of the erythropoietin hormone**.
- 3- **Store** of iron, copper and Vit.B₁₂ .
- 4- Liver **synthesis RBCs in the fetal stage** (before the formation of the bone) and can synthesis RBCs temporally in case of bone marrow destruction.

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6) Bone marrow:

It is the **factory of RBCs production**. When it is destroyed (e.g. by X- rays, radium, atomic irradiation and some toxins), all types of blood cells are not formed which leads to **aplastic anemia**.

FACTORS AFFECTING

A) Physiological:-

- 1) **Species:** birds 3, rabbit 4, man 5, dog, cat, pig & cattle 6-8; horse & sheep 7-13 ($\times 10^6 / \text{mm}^3$).
- 2) **Breed:** e.g. light horse (11) while, draft horse ($8 \times 10^6 / \text{mm}^3$).
- 3) **Individual:** it differs according to genetic performance, nutritional state, health.....

4) **Age:** newly born (till about 3-4 months old age) has greater number of RBCs (due to slight hypoxia that occurs during pregnancy), which have very important function for supplying the body of newly born with iron (from destructed RBCs), to compensate the decreased level of iron in the milk of mother.

Then the number decreased under the normal number of the adult. Then, gradually increases till the age of maturity, then decline again in old age.



FACTORS AFFECTING

5) **Sex:** male has an excess RBCs number than female due to:-

- 1- Testosterone in male.
 - 2- Cyclic bleeding in female (menses in primates, metestrus bleeding in cow, proestrus & estrus bleeding in bitch & queen).
 - 3- Bleeding during parturition.
 - 4- Decreased iron level in females due to cyclic bleeding, pregnancy and lactation.
- 6; 7;8) Hot weather; muscle exercise; emotion states: they lead to release of epinephrine, which contract the splenic capsule that squeeze spleen leads .

FACTORS AFFECTING

9) Oxygen pressure:-

1- **Decrease oxygen pressure (as in high altitude):** stimulate RBCs production from the bone marrow and also increase the release of the splenic storage of RBCs (by contraction of the splenic capsule).

2- **Increase oxygen pressure** (as living or working under the sea level e.g. workers in deep mines), leads to decrease RBCs production.

10) **Sleep:** leads to decrease RBCS number.

FACTORS AFFECTING

B) Pathological:-

- 1) That **decreases** the RBCs number: as in different types of **anemias**.
- 2) That **increases** the RBCs number (**Polycythemia**): as in the following cases:-
 - a- **Emphysema**: leads to decrease alveolar surface, which cause hypoxia, that increase RBCs number.
 - b- **Congestive heart failure**: leads to stagnation of systemic circulation, which cause hypoxia, that increase RBCs number.
 - c- **Co- poisoning**: leads to formation of carboxy-Hb, which cause hypoxia, that increase RBCs number.

HAEMOGLOBIN(HB)

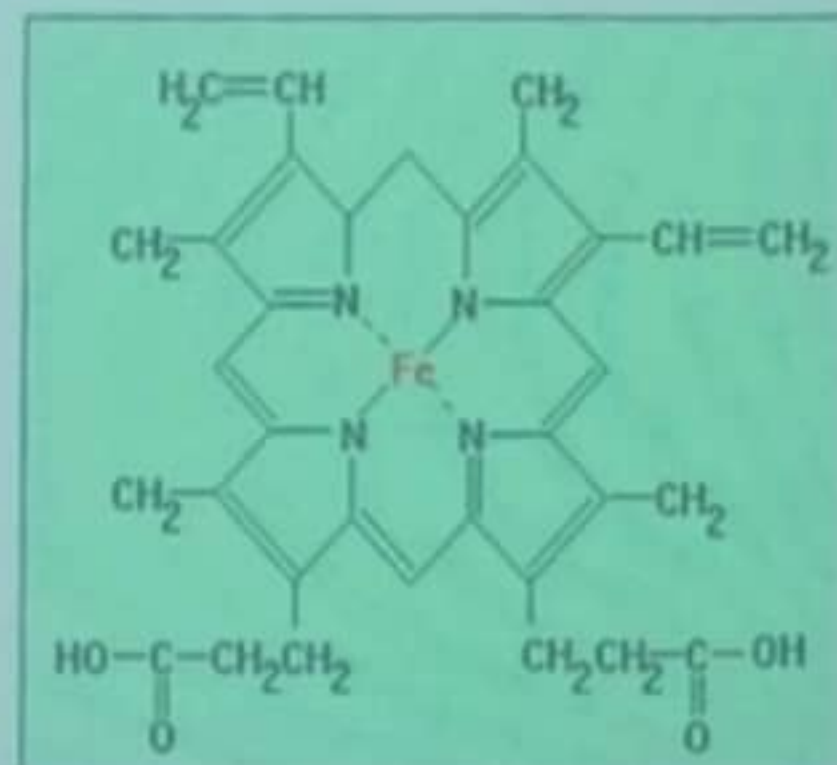
- Hb is the **red pigment** of RBCs and blood. Its biosynthesis starts in the stage of erythrocytoblast in the red bone marrow and continues in the subsequent stages of erythropoiesis. Hb is a complex iron containing **conjugated protein**, composed of pigment and simple protein.

Structure of Hb:-

- 1) Globin: compose about **96%** of the Hb molecule.
 - It is a simple **protein** consists of several amino acids(about 186).
 - Globin molecule is **species different** and considered as a haem carrier.

2) Haem: compose about 4% of the Hb molecule.

- Each Hb molecule containing **4 haem**. Haem structure is **identical** in all species.
- It carries the **oxygen** and a great part of CO_2 . It gives the blood its **red color**.
- The amino acid glycine + Acetate $\rightarrow$ Pyrrole ring.
- 4 Pyrrole ring $\rightarrow$ Protoporphyrine chain.
- Protoporphyrine + Iron (Fe^{+2}) $\rightarrow$ Haem.
- 4 Haem + Globin $\rightarrow$ Hb.



HAEMOGLOBIN(HB)

Structure of haem molecule:

Iron compose about **0.334%** of Hb, bind haem with globin and able to carry and give O_2 . The body conserves iron resulted from Hb destruction, so it's dietary need for HB synthesis is small.

When O_2 carry on the Hb converted it to **oxy-Hb (HbO_2)** and when the Hb give its O_2 , called **reduce Hb** or **de-oxy Hb**.

TYPES OF HB

A) Developmental Hb:-

- 1) Embryonic Hb (HbP): It is formed in the RBCs of the embryo in the 1st trimester of pregnancy.
- 2) Foetal Hb (HbF): It is formed in the fetus RBCs in the 2nd trimester of pregnancy substituted the HbP, then gradually decrease and substituted by the adult Hb and finished from the body at 4 months age.
 - It has a greater affinity to the oxygen than the adult Hb but the RBCs containing HbF have more affinity to hemolysis.

- If the HbF stay in the RBCs after the age 4 months, it causes a disease called **Thalassemia** (*Mediterranean anemia*). It is a genetic disease characterized by: hemolysis of the RBCs; precipitation of the iron in the tissues; reduction in the body immunity and death. Treatment:

1. blood transfusion.
 2. Removal of the excess iron from the body by using a drug.
1. Increase the immunity.



3) Adult Hb (HbA): It is formed in the fetus RBCs in the 2nd trimester of pregnancy with the HbF, and then gradually substituted it. At 4 months of age all the HbF are substituted by HbA and all the RBCs in the body containing HbA till the end of life.

B) Functional Hb:-

- 1) Oxy Hb (HB O₂), when the Hb carries O₂.
- 2) Deoxy or reduce Hb, when the Hb give its O₂.
- 3) Carbamino-Hb, when the Hb carries CO₂.

C) Abnormal (pathological) Hb:-

1) Met-Hb:-

- It is a stable combination of Hb with O_2 (oxidation not oxygenation, i.e. the ferrous ions of Hb transformed to ferric ions), due to exposure to strong oxidizing agents as **ferric cyanide, potassium permanganate, nitrite, sulfonamides and potassium chloride**. It gives the blood a **dark brown** color.
- Met-Hb is a **non-toxic** compound formed in a small amount in the circulation. Reducing compounds in the RBCs as **Vit.C** and **glutathione** are preventing its accumulation. But in cases of accumulation of Met-Hb may be lead to **asphyxia** and death due to the Hb fails to give its oxygen to the tissues.

TYPES OF HB

2) CarboxyHb (HbCo):-

- It is a **bright red color**, which mislead in thinking that the body had abundant oxygen. HbCo is a compound formed when the Hb molecule carry **carbon monoxide (coal gas, Co)**. The affinity of Hb for Co is more than 200 times greater than for oxygen. So, even 0.1% Co in the inhaled air leads to HbCo formation which is fatal.
- Weak poisoning with coal gas is a reversible process. With inhalation of fresh air, Co splits off gradually from the Hb and is eliminated. Breathing of pure oxygen gives 20 fold increases in the rate of HbCo splitting. With severe poisoning, it is necessary to apply artificial respiration.

3) Sulfo-Hb:-

The Hb molecule combined with **sulfo group** as in cabbage, turnip and anion. It is a **reversible process**. It gives the blood a **dark brown color** and may cause brown patches on the skin.

MYOGLOBIN

- It is a **muscular Hb** that occurs in skeletal and cardiac muscles. It is composed of globin and one molecule only of haem. It is **more trapping** and more retaining to oxygen than Hb.
- Myoglobin can carry about 14% of all the body oxygen. This property is important for the supply of oxygen to the working muscles. When blood flow is arrested in some parts of the muscle through compression of the capillaries during muscle contraction, an oxygen supply is maintained for some time to its fibers by the oxygen combined with myoglobin (so it is called oxygen store in the body).
- During rest and muscle relaxation (myoglobin not gives its oxygen), the muscle takes its need of oxygen from the blood Hb.

LIFE SPAN OF ERYTHROCYTE

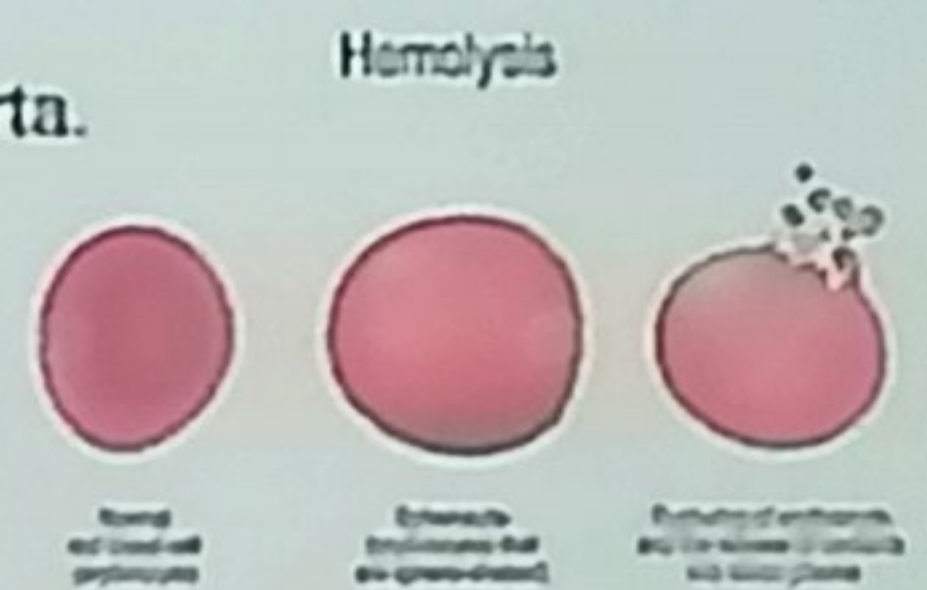
- ❖ The life span of RBCs in man and most of the domestic animals is about 120 days (50 days in rats and rabbits, 25 days in mice and poultry).
- ❖ Therefore, about $1/120$ of the RBCs number is destroyed daily and the bone marrow compensate that by forming the same number of the RBCs ($1/120$) daily.

HEMOLYSIS (DESTRUCTION OF THE ERYTHROCYTES)

- RBCs are always **destroyed** in the circulation and engulfed by the **reticuloendothelial cells** (Tissue macrophages) in different organs especially in the **bone marrow, liver and spleen**.

Causes of hemolysis:-

- 1- Frequent forcible ejection of RBCs by the heart into aorta.
- 2- Agglutination (their collection) of the RBCs.
- 3- Their frequent squeezing in the fine capillaries.
- 4- Senility of the RBCs.
- 5- Hereditary diseases as **Thalassemia, Favism and Sickle's cell anemia**.
- 6- Pathological as in some toxins, parasite, bacteria and chemicals.



FATE OF RBCS

1. The **reticuloendothelial cells** after phagocytized the hemolyzed RBC, hydrolyzed the Hb molecule to globin and Haem.
2. Then, the **globin** hydrolyzed to **amino acids** which reused again.
3. The **haem** hydrolyzed to **iron** (reused again, about 25 mg daily) and **biliverdin**.
4. **Biliverdin** (green in color) reduced to **bilirubin** (yellow in color).
5. Bilirubin is a **toxic and filterable** substance, so it carry with a plasma protein to form Haemobilirubin and transfer to the **liver**.

6. **Bilirubin** (in the liver) + **Glucuronic acid or sulphur** → Cholebilirubin (excreted with bile) → Stercobilinogen (in the large intestine) → eliminated with faeces → Stercobilin (after oxidation by air).
7. Part of the **stercobilinogen** → Kidney (through general circulation) → Urobilinogen (in the urine) → Urobilin.
8. Other part of the **stercobilinogen** goes to general circulation then to liver → bile.....

